

PEA305:ANALYTICAL SKILLS-I

L:2 T:1 P:0 Credits:3

Course Outcomes: Through this course students should be able to

CO1 :: apply the procedural fluency with number system and mathematical operations to solve the stated problems

CO2 :: determine an appropriate approach to solve problems related to percentage and profit & loss

CO3 :: analyze the data given and interpret the number and alphanumeric series

CO4 :: use the analytical concepts to solve the questions of ratio and proportion

CO5 :: apply the concepts of permutation, combination and probability to solve related problems

CO6 :: analyze the reasoning aptitude problems such as blood relation and direction sense to solve related problems

Unit I

Vedic maths : fast calculation techniques

Number system : classification of numbers, divisibility rules, factorials, unit digit calculation, number of factors, remainder theorems, HCF & LCM

Average : average calculations, inclusion and exclusion, weighted average

Unit II

Percentage : percentage comparison, calculation of percentages, percentage to fraction conversion, successive percentage change, problems based on marks, election, population and percentage error

Profit loss discount : calculations of profit and loss percentage, concept of cost price, selling price and marked price, problems on discount, faulty weights, number of articles, and common selling price

Unit III

Logical reasoning : number series, alphabet series, alphanumeric series, alphabet test, coding and decoding, language coding

Unit IV

Ratio and proportions : concept of ratio and proportion, problems based on ages and partnership

Alligation and mixtures : concept based question of alligation and mixtures, replacement based questions

Unit V

Permutation and combination : principles of counting, problems based on selection, arrangement, numbers, words, rank and distribution, geometric application, circular arrangement

Probability : concept of probability, classification of events, problems based on coins dices and cards, conditional probability

Unit VI

Simple and compound interest : comparison of simple and compound interest, concept of interest calculation, compounding based problems on depreciation and growth

Analytical reasoning : blood relations, direction sense test

Text Books:

1. ANALYTICAL SKILLS-I by S. CHAND EXPERTS, S Chand Publishing

References:

1. QUANTITATIVE APTITUDE FOR COMPETITIVE EXAMINATIONS by DR.R.S. AGGARWAL, S Chand Publishing
2. A MODERN APPROACH TO VERBAL AND NON-VERBAL REASONING by DR. R.S. AGGARWAL, S Chand Publishing

